FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject	:	**System Analysis and Design (SECP2613)**
Section	:	**01, 02**
Task	:	**Analysis and Planning**
Phase 1	:	**Project Proposal and Planning (12%)**
Phase 2	:	**Information System Gathering and Requirement (12%)**
Due	:	**24.5.2024**

	Task	Deliverables (items in Proposal)
1	Get information on the current systems ***(AS-IS)***	Methods used for gathering information ● Details of the information gathering method (interview questions). ● Summary from the method (interview sample). ● Document Analysis: collect the exiting documents related to the current system ● Requirement analysis based on interview conducted
2	Conduct interactive and unobtrusive methods to gather the information from the relevant stakeholders	
3	Observation	● Takes notes from the existing software (you can understand how it works and what users like and dislike about it). ● Example https://edaftar.skm.gov.my/, Sistem Pendaftaran Koperasi (eDAFTAR) yang dibangunkan ini adalah untuk membantu Suruhanjaya Koperasi Malaysia mengurus dan memantau semua transaksi yang berkaitan Koperasi khususnya Pendaftaran dan Pembatalan Koperasi.
4	Conduct problem background ● Organizational background study, ● What areas the systems to cover (scope) ○ What will be included ○ What will not be included ● Benefits from the systems/contribution	● Introduction (Overview of the Project) ● Problem Statement ● Proposed Solutions ● Objectives ● Scope of the Project
5	Propose feasibility study (technical, operational, economical)	● Findings from feasibility study (background study) ● Cost Benefit Analysis (CBA)

6	Project Plan ● Time to produce the systems ● Who will produce it ● A deadline (expected completion date)	● Work based Structure (WBS) ● Pert Chart ● Gantt Chart
7	Project Management Tools	**Please refers to the explanation in third page.*
8	Gathers information about what the users want in the systems ***(AS-IS)***	● Current business process/workflow ● Identification of functional and non-functional requirement (AS-IS). ● Logical DFD (AS-IS) system (based on current business process/workflow) 1. Context Diagram 2. Diagram 0 3. Child Diagram (if applicable)

Prepare a proposal with the items below:

1.0 Introduction
2.0 Information gathering process
  2.1 Method used
  2.2 Summary from method used (include example Interview/Questionnaire/Observation)
3.0 Background Study
4.0 Problem Statement
5.0 Proposed Solutions (include feasibility study – technical, operational, economical - CBA)
6.0 Objectives
7.0 Scope of the Project
8.0 Project Planning
  8.1 Human Resource
  8.2 Work Breakdown Structure (WBS)
  8.3 PERT Chart (based on WBS)
  8.4 Gantt Chart
9.0 Requirement Analysis (based on AS-IS analysis)
  9.1 Current business process (scenarios, workflow)
  9.2 Functional Requirement (input, process and output)
  9.3 Non-functional Requirement (performance and control)
  9.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)
10.0 Benefit and Overall Summary of Proposed System

Project Management Explanation

- **GitHub Account Requirement**: All students must have a GitHub account to participate in the project. This ensures that each team member can contribute to the repository and participate in the version control and project management processes.
- **Repository Creation and Team Setup**:
    - The group leader is responsible for creating a repository named "GroupName_Project1_SAD_20232024". This will be the central repository where all project work is stored and managed.
    - The group leader must invite all team members to join the repository, ensuring everyone can contribute to the project. This collaborative environment is where the bulk of your project development will take place.
    - Share the repository ID with the lecturer, enabling them to join the repository and monitor the project's progress. This allows for real-time feedback and guidance throughout the project lifecycle.
- **Understanding Forking:**
    - A "fork" is essentially a personal copy of someone else's project, allowing you to experiment with changes without affecting the original project. Although your main project work will happen in the shared repository, understanding forking is crucial for several reasons:
        - **Experimentation**: You might want to fork the project repository to experiment with new features or changes independently. This allows for innovation without risking the integrity of the main project.
        - **Contribution Workflow**: Understanding forking is essential for contributing to open-source projects or any project outside your direct control in the future. You can fork a project, make your improvements, and then propose these changes to the original project through a pull request.
        - **Learning Version Control**: Forking introduces you to advanced version control practices, enhancing your GitHub skills and preparing you for collaborative software development.

Inside the report, please include the following:

- **URL of the GitHub Repository**: Provide the URL of the GitHub repository where the project code, documentation, and management artifacts are stored.
- **Repository Snapshot**: Submit a snapshot or screenshot of the repository to demonstrate the current state of the project.
- **Kanban Board Integration**: Create a Kanban board within GitHub Projects for task management and workflow visualization.
    - Task Creation: Break down the project into manageable tasks and create a card for each task on the Kanban board. Ensure tasks are clearly defined and assigned to team members.
    - Columns Setup: Organize the Kanban board into columns that represent the workflow stages, such as To Do, In Progress, Review, and Done.
    - Progress Tracking: Regularly update the Kanban board to reflect the current status of each task. Move cards across columns as work progresses.
    - Meeting Milestones: Use the board to help the team meet milestones by visually tracking progress and identifying bottlenecks early.
- **Version Control Practice**s: Emphasize the importance of using version control best practices, including feature branching, pull requests, and code reviews.

Phase 1 Mark Allocation

	ITEM	%
PLO2 (Apply)	Exercises related to CLO3	2
	Technical aspect for CLO3 (Part B)	3
PLO3 (Problem Solving)	Project plan aspect for CLO3 (Part C)	5
PLO7 (Teamwork)	Self/Peer Evaluation	1
	Instructor Observation (Part A)	1
	TOTAL	**12**

PART A (1%) Teamwork				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Teamwork	The team worked **WELL TOGETHER** to achieve objectives. Each member contributed in a valuable way to the project.	The team worked **WELL TOGETHER MOST OF THE TIME**, with only a few occurrences of communication breakdown. Members were mostly respectful of each other.	The team worked together **most of the time**, **POSSIBLE FAILURE** to collaborate when appropriate. Some members would work independently, without regard to objectives or priorities.	Team did **NOT COLLABORATE** or communicate well.
PART B (3%) Technical Aspects				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Explains Issues and Clear Content Organization	**CLEARLY** identifies and summarizes main issues and successfully explain why/how they are problems or questions; and identifies embedded or implicit issues, addressing their relationship	**SUCCESSFULLY** identifies and summarizes the main issues, but does not explain why/how they are problems or create questions.	**IDENTIFIES** main issues but does not summarize or explain them clearly or sufficiently	**FAILS** to identify, summarize or explain the main problem or questions. Represents the issues inaccurately or inappropriately.
	Method of organization is **WELL-SUITED** to assignment; clear intro, body, and conclusion with effective transitions.	Organization **SUPPORTS** assignment and purpose; sequence of ideas could be improved	**SOME SIGNS** of logical organization. May have abrupt or illogical shifts and ineffective flows of ideas	**POORLY** organized OR demonstrates serious problems with progression of ideas; a written form of speech

Content Development & Writing	Content is somewhat **ACCURATE AND CLEAR**; offers **SOLID AND ACCURATE** reasoning; contains appropriate details and / or examples	Content is somewhat **ACCURATE AND FAIRLY CLEAR**; offers **SOLID BUT LESS ACCURATE** reasoning; contains some appropriate details and / or examples	Content is somewhat **VAGUE** OR only **LOOSELY RELATED** to the writing task; at times may be **OFF TOPIC** OR **TOO BROAD** with limited support	Content **UNCLEAR; LAPSES IN COHERENCE** OR **NO RELATION** to writing task; offers simplistic, undeveloped support for ideas.
	Written works has **NO MAJOR ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization.	Written works is **RELATIVELY FREE OF ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization	Written work has **SEVERAL MAJOR ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization	Written work has **SERIOUS AND PERSISTENT ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization
PART C (5%) Project Plan Aspect				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Complete Project Plan	Complete, precise and clear task description. Time allocation for each task is **HIGHLY ACCEPTABLE**.	Task description is **FAIRLY COMPLETE AND CLEAR**. Time allocation for each task is **ACCEPTABLE**.	Task description is vague and unclear. Time allocation for each task is **FAIRLY ACCEPTABLE**.	Unclear and incomplete task description. Time allocation for each task is **UNLOGIC** and **UNACCEPTABLE**.
Complete Budget Planning	Complete and clear budget allocation. Budget allocation is **HIGHLY ACCEPTABLE**.	Budget description is **FAIRLY COMPLETE AND CLEAR**. Budget allocation is **ACCEPTABLE**.	Budget description is vague and unclear. Budget allocation is **FAIRLY ACCEPTABLE**.	Unclear and incomplete budget description. Budget allocation for each task is **UNLOGIC** and **UNACCEPTABLE**.
Complete Task and Human Resource Planning	Human resource for each task is **HIGHLY ACCEPTABLE**.	Human resource in task allocation is **ACCEPTABLE**.	Human resource in task allocation is **FAIRLY ACCEPTABLE**.	Unclear and incomplete task description. Time allocation for each task is **UNLOGIC** and **UNACCEPTABLE**.

Phase 2 Mark Allocation

	ITEM	%
PLO2 (Apply)	Exercises related to CLO3	2
	Technical aspect for CLO3 (Part B)	3
PLO3 (Problem Solving)	Requirement aspect for CLO3 (Part C)	5
PLO7 (Teamwork)	Self/Peer Evaluation	1
	Instructor Observation (Part A)	1
	TOTAL	**12**

PHASE 2 RUBRIC

note: Rubric is taken from Assessment for Learning Handbook, Tools for Assessment and MIS 5302: Case Study Analysis Rubric.

PART A (1%) Teamwork				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Teamwork	The team worked **WELL TOGETHER** to achieve objectives. Each member contributed in a valuable way to the project.	The team worked **WELL TOGETHER MOST OF THE TIME**, with only a few occurrences of communication breakdown. Members were mostly respectful of each other.	The team worked together **most of the time**, **POSSIBLE FAILURE** to collaborate when appropriate. Some members would work independently, without regard to objectives or priorities.	Team did **NOT COLLABORATE** or communicate well.
Part B (3%) Technical Aspects				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Organizational Analysis (Able to identify key information/data from relevant sources)	Displays a **COMPREHENSIVE UNDERSTANDING** of an organization's structure, culture and available resources as well as its key advantages and distinctive attributes. Provides **INSIGHTFUL ANALYSIS** supported by data.	**PRECISELY** identifies and displays an understanding of an organization's structure, culture and available resources, as well as its key advantages. Provides **ORIGINAL ANALYSIS** with some supporting data.	**CLEARLY** identifies an organization's structure, culture and available resources. Moves beyond reporting of facts to offer some **ADDITIONAL ANALYSIS**.	**LIMITED** ability to identify and analyze an organization's structure, culture and available resources. Mostly reports facts with **LITTLE OR NO ANALYSIS**.
Stakeholder Perspectives	Demonstrates ability to identify **ALL** relevant stakeholders; to identify and reflect on their perspectives and interests as they relate to a particular business issue; and to **MAP FULLY THE INTERACTIONS** between stakeholders.	Demonstrates ability to identify **MOST** relevant stakeholders; to identify and reflect on their perspectives and interests as they relate to a particular business issue; and to begin to **MAP THE INTERACTIONS** between stakeholders.	Demonstrates ability to identify **SOME** relevant stakeholders. Displays **SOME** ability to identify relevant stakeholders and their perspectives and interests as they relate to a particular business issue.	Demonstrates **LIMITED OR NO** ability to identify relevant stakeholders and their perspectives and interests as they relate to a particular business issue.

Business StoryTelling (Able makes connections between ideas/facts/data to definition of requirements)	**IMMEDIATELY GRABS** attention. **CLEARLY** establishes the appropriate context for why information is important. "Story" is focused and presented in a **MEMORABLE** fashion.	**GRABS ATTENTION**. Establishes the appropriate context for why information is important. "Story" is **CLEAR AND FOCUSED**.	Does **NOT IMMEDIATELY GRAB** attention. Somewhat establishes the appropriate context for why information is important. "Story" is **SLIGHTLY DIFFUSE AND UNFOCUSED**.	**STRUGGLES** to grab attention throughout. Little or no establishment of appropriate context for why information is important. "Story" is **DIFFUSE AND UNFOCUSED**
Supporting Evidence	**Evidence** presented to support argument/recommendation is **STRONG AND CREDIBLE**, exhibiting depth of idea. Counter evidence is **ADDRESSED AND SUFFICIENTLY** refuted.	**Evidence** is **SUFFICIENT** to support argument/recommendation. Counter evidence is **ADDRESSED BUT NOT SUFFICIENTLY** refuted.	**Evidence** is **SOMEWHAT SUFFICIENT** to support argument/recommendation. **IGNORES SOME** of the counter evidence to argument.	Evidence is **INSUFFICIENT OR MISINTERPRETED** in support of argument/recommendation. **IGNORES ANY** counter evidence to argument.
Content Development & Writing	Content is somewhat **ACCURATE AND CLEAR**; offers **SOLID AND ACCURATE** reasoning; contains appropriate details and / or examples	Content is somewhat **ACCURATE AND FAIRLY CLEAR**; offers **SOLID BUT LESS ACCURATE** reasoning; contains some appropriate details and / or examples	Content is somewhat **VAGUE** OR only **LOOSELY RELATED** to the writing task; at times may be **OFF TOPIC** OR **TOO BROAD** with limited support	Content **UNCLEAR; LAPSES IN COHERENCE** OR **NO RELATION** to writing task; offers simplistic, undeveloped support for ideas.
	Written works has **NO MAJOR ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization.	Written works is **RELATIVELY FREE OF ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization	Written work has **SEVERAL MAJOR ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization	Written work has **SERIOUS AND PERSISTENT ERRORS** in word selection and use, sentence structure, spelling, punctuation and capitalization
Part C (5%) Requirement Aspects				
Criteria	**High (4)**	**Average (3)**	**Low (2)**	**Unsatisfied (1)**
Identify the requirement based from the case study.	Correctly interpreted **ALL** business rules with **CLEAR** idea on the input, process and output.	Correctly interpreted **MOST** of the business rules with **GOOD** idea on the input, process and output.	I**NACCURATELY** interpreted the business rules with **WRONG** idea on the input, process and output.	**UNABLE** to interpret the business rules with **NO IDEA** on the input, process and output.

Produce a context diagram for the current/manual system based from the case study.	**GOOD** understanding of the requirement and name of the main process, entities and data flows are **ACCURATE** and **SUITABLE**.	**PARTIAL** understanding of the requirement and have **MINIMAL ERROR** in naming the main process, entities and data flows.	**INCORRECTLY** understand the requirement and put **INACCURATE** name for the main process, entities and data flows.	**FAIL** to understand the requirement and put **WRONG** name for the main process, entities and data flows.
Produce Diagram 0 and (possible child diagram) for the context diagram.	**ACCURATE** use of names in entities, processes, data flows and data store – **ALL** names are appropriately used in regard of its requirement.	Some **MINOR** errors in the use of names in entities, processes, data flows and data store – **FEW** names are not clear/suitable with its requirement.	**INACCURATE** use of names in entities, processes, data flows and data store – the name **DID NOT** correlate with its requirement.	**WRONG** use of names in entities, processes, data flows and data store. Shows **NO** understanding in identifying the entities, processes, data flows and data store.